CSV-KPI vs HDF-KPI accuracy diff — AshLey 0011 log

gen7v2 CSV scripts vs UDP_KPI (HDF) · IN/OUT pair 0011_b05 + rR00060100 · generated 2026-09-22 13:27

1 · What was run

Pipeline	Input	Sensors	Time	Peak RAM
CSV gen7v2 detection script	18 CSVs (DET 18MB + RDD 106MB + VSE per side/sensor)	FC/FL/RL	~33/39/39 s	single process, GB-class
UDP_KPI HDF (ours)	hdf_11 pair (518/504 MB)	FC/FL/RL	~2–3 s/sensor	164–735 MB, one process/sensor, 3 GB self-cap

2 · Measured headlines (detection)

Sensor	CSV result4 (r,v,t,p)	CSV result3 (r,v)	HDF hero (comparable)	HDF overall
FC	88156/88160 = 100.0%	100.0%	85820/85864 = 99.95%	same
FL	115979/115985 = 99.99%	100.0%	113160/113277 = 99.90%	same
RL	119163/119167 = 100.0%	100.0%	116460/116606 = 99.87%	same

Bottom line up front: on this log both pipelines converge to ~100% (residual 0.05–0.13 pp). The resim reproduces detections almost bit-identically, so every gate choice saturates. The accuracy *difference* between the pipelines lives in the method, and only shows on divergent logs (see §5).

3 · ELI13: why two calculators can disagree

Two teachers grade the same replay. The CSV teacher is **extremely strict**: a distance must match within 1 centimeter, speed within 1.5 cm/s, direction within half a degree — like demanding a traced drawing match to the width of a hair. The HDF teacher is **lenient**: 2.5 m, 2.5 m/s, 143 degrees — like accepting anything in the same room. On this log the replay is so perfect that **both teachers give ~100%**. On a sloppy replay (AL_0004), the strict teacher fails almost everything while the lenient one still passes a quarter — same photos, different rulers. They also count photos slightly differently (which scans enter the denominator), so even identical answers differ in the second decimal.

4 · PhD: method face-off

Axis	CSV gen7v2	UDP_KPI (HDF)
Layout	Wide CSV: af_dets_{i}_* columns (FC 768, FL/RL 544 present of 680)	Tall HDF: (scans × 6000) datasets parsed per scan
Gates	ran 0.01 m, vel 0.015 m/s, theta/phi 0.00873 rad	ran/vel/theta/phi 2.5 (m, m/s, rad, rad) — ~250× looser angles
Prefilter	exact (rdd1_rindx, rdd1_dindx) equality, one-to-one consume	same prefilter, same discipline
Scan scope	drops scan_index==0, num_af_det==0, dedups; merges DET∩RDD(∩CDC); 391/392 merged scans	keeps all scans, common-scan alignment; 384–385 common scans
Scan identity	DET-list scan_index column	Stream_HDR scan_index (sane-checked)

Axis	CSV gen7v2	UDP_KPI (HDF)
Clipping	effective af width = min(veh,sim) present columns	native HDF width; sim count capped by veh count (known quirk)
RDD values	raw P21 ints ×4.768371582e-07 before compare (verified same raw ints as HDF int32)	rdd2 never numerically compared (prefilter uses rdd1 indices; matrix uses DET floats) — no unit bug either side
Reported	result3 subset (r,v) + result4 full (r,v,t,p); scan-yield/mileage tables	hero (comparable) + overall (coverage-inclusive); per-scan table

5 · Mechanism demo: gate pass rates on RDD-matched pairs (FL)

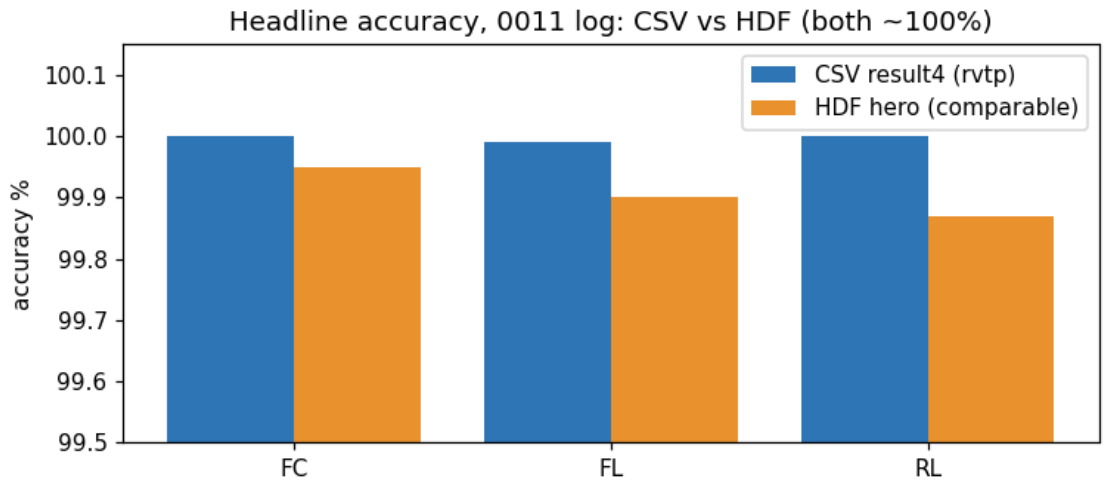
Signal	0011 pass @CSV gates	0011 pass @our gates	0011 max |delta|
ran	99.38%	99.95%	138.0 m
vel	99.38%	99.93%	29.7 m/s
theta	99.36%	100.00%	2.04 rad
phi	99.42%	100.00%	0.69 rad

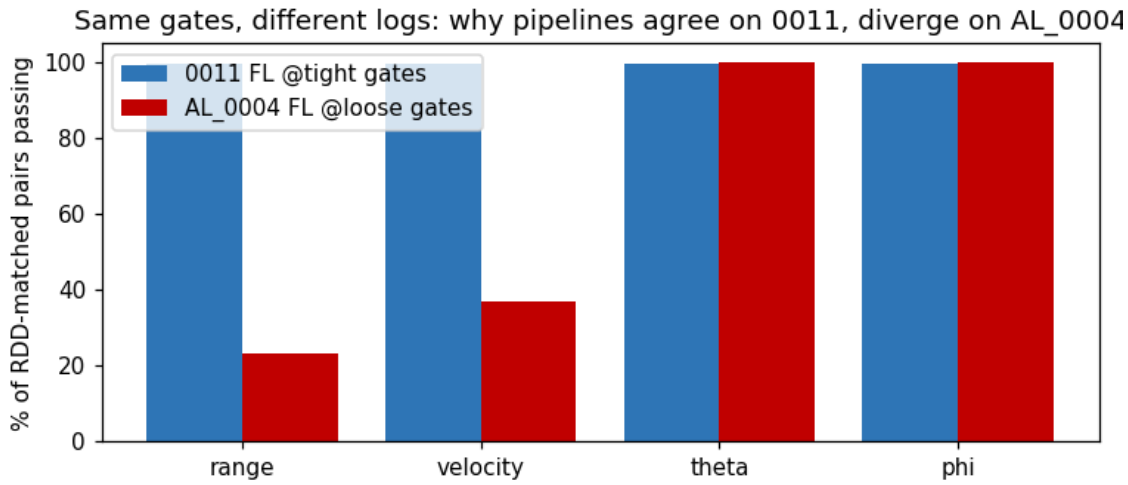
113,277 RDD-matched pairs: at the CSV microscope ~99.4% pass each signal; at our gates ~99.9%. That 0.5 pp is exactly the measured headline gap (99.99 vs 99.90). Contrast AL_0004 FL at *our loose* gates: ran 23.4%, vel 37.1%, theta/phi ~100% — at CSV gates it would read ≈0%. Same rulers, different replays: direction-consistent but range/velocity-divergent traffic (delta spread p50 10 m → p99 ~100 m rules out constant bias or unit slip).

6 · Residual 0.05–0.13 pp on 0011 (both saturated — from scope, not gates)

CSV denominators (88160/115985/119167) exceed ours (85864/113277/116606) by ~2.5%: different scan sets (391/392 merged vs 384/385 common — header vs list-property scan identity + zero/dedup filtering), different width clipping (544 present vs native), and our sim-count-capped-by-veh quirk. Per-scan CSV warnings confirm robustness handling (544-vs-680 columns) on both sides.

7 · Figures





Left: headlines converge (~100% both). Right: pass-rate fingerprint — 0011 passes even the microscope; AL_0004 fails range/velocity even under the lenient gates.

8 · Recommendations

1) Quote CSV result4 and HDF hero side by side only with gate context: strict ≈ spec compliance, lenient ≈ scene similarity. 2) For parity studies, tighten our gates toward 0.01/0.015/0.00873 (config-only change) rather than loosening theirs. 3) For divergent logs, lead with the per-signal pass fingerprint (angles vs range/velocity) before any headline — it separates replay fidelity from ruler strictness. 4) CSV outputs (report + debug txt) and our HTML/sidecars for log 0011 all live in out_csv_hdf/ beside this PDF.